

Dot Product (Scalar product)

$$a = a_1 \underline{i} + a_2 \underline{j} + a_3 \underline{k} \quad b = b_1 \underline{i} + b_2 \underline{j} + b_3 \underline{k}$$

$$a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Note: If $a = 0$ or $b = 0$, then $a \cdot b = 0$.

Example.

$$a = \underline{i} - 2\underline{j} + 3\underline{k} \quad b = -2\underline{i} + 3\underline{j} + 4\underline{k}$$

1. $a \cdot b$ 2. $a \cdot a$

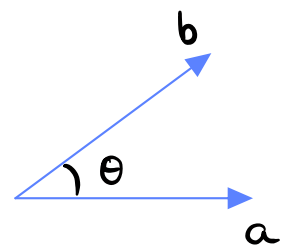
Sol. 1. $a \cdot b = (\underline{i} - 2\underline{j} + 3\underline{k}) \cdot (-2\underline{i} + 3\underline{j} + 4\underline{k})$
 $= 1(-2) + (-2)(3) + (3)(4)$
 $= 4$

2. $a \cdot a = (\underline{i} - 2\underline{j} + 3\underline{k}) \cdot (\underline{i} - 2\underline{j} + 3\underline{k})$
 $= 1^2 + 2^2 + 3^2$
 $= 14$

Geometric description of scalar product

For vectors a and b , we have

$$a \cdot b = |a| |b| \cos \theta$$



θ is the angle between a and b

Note: $0^\circ \leq \theta \leq 180^\circ$

Example.

If $|a| = 4$, $|b| = 5$, the angle between a and b is 30° , find $a \cdot b$.

sol. $a \cdot b = |a||b| \cos 30^\circ$
 $= 4 \times 5 \times \frac{\sqrt{3}}{2}$
 $= 10\sqrt{3}$

Properties

* $a \cdot b = b \cdot a$

* $k(a \cdot b) = (ka) \cdot b = a \cdot (kb)$

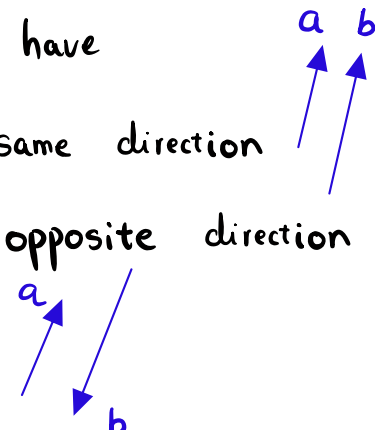
* $a \cdot 0 = 0$

* $a \cdot (b + c) = a \cdot b + a \cdot c$

* If the vectors a and b are perpendicular, then $a \cdot b = 0$

* If $a \cdot b = 0$ for **non-zero** vectors a and b , then the vectors a and b are perpendicular.

* For parallel vectors a and b , we have

$$a \cdot b = \begin{cases} |a||b|, & a \text{ and } b \text{ are parallel in the same direction} \\ -|a||b|, & a \text{ and } b \text{ are parallel in the opposite direction} \end{cases}$$


* For unit vectors $\underline{i}$, $\underline{j}$ and $\underline{k}$, we have

$$\cdot \underline{i} \cdot \underline{i} = \underline{j} \cdot \underline{j} = \underline{k} \cdot \underline{k} = 1$$

$$\cdot \underline{i} \cdot \underline{j} = \underline{i} \cdot \underline{k} = \underline{j} \cdot \underline{k} = 0$$

Example 21

a Simplify $a \cdot (b + c) - b \cdot (a - c)$.

b Expand the following:

i $(a + b) \cdot (a + b)$ **ii** $(a + b) \cdot (a - b)$

Solution.

$$\begin{aligned} \text{(a). } a \cdot (b + c) - b \cdot (a - c) &= a \cdot b + a \cdot c - b \cdot a + b \cdot c \\ &= a \cdot c + b \cdot c \end{aligned}$$

$$\begin{aligned} \text{(b). i. } (a + b) \cdot (a + b) &= a \cdot a + a \cdot b + b \cdot a + b \cdot b \\ &= a \cdot a + 2 \cdot a \cdot b + b \cdot b \end{aligned}$$

$$\begin{aligned} \text{ii. } (a + b) \cdot (a - b) &= a \cdot a - a \cdot b + b \cdot a - b \cdot b \\ &= a \cdot a - b \cdot b \end{aligned}$$

Example 22

Solve the equation $(i + j - k) \cdot (3i - xj + 2k) = 4$ for x .

Solution. $(i + j - k) \cdot (3i - xj + 2k) = 4$

$$3 - x - 2 = 4$$

$$1 - x = 4$$

$$x = -3$$

Finding the magnitude of the angle between two vectors

$$a \cdot b = |a||b| \cos \theta$$

$$a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\Rightarrow \cos \theta = \frac{a \cdot b}{|a||b|} = \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{|a||b|}$$

Example 23

A , B and C are points defined by the position vectors a , b and c respectively, where

$$a = i + 3j - k, \quad b = 2i + j \quad \text{and} \quad c = i - 2j - 2k$$

Find the magnitude of $\angle ABC$, correct to one decimal place.

$\angle ABC$ is the angle between vectors

$\vec{BA}$ and $\vec{BC}$.

$$\vec{BA} = a - b = -i + 2j - k$$

$$\vec{BC} = c - b = -i - 3j - 2k$$

$$\vec{BA} \cdot \vec{BC} = |\vec{BA}| \cdot |\vec{BC}| \cos(\angle ABC)$$

$$|\vec{BA}| = \sqrt{1+2^2+1^2} = \sqrt{6}$$

$$|\vec{BC}| = \sqrt{1+9+4} = \sqrt{14}$$

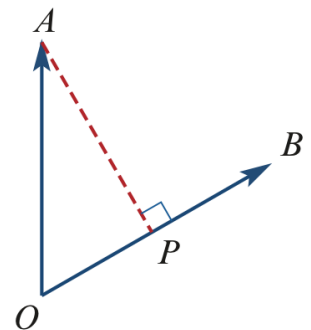
$$\vec{BA} \cdot \vec{BC} = 1 - 6 + 2 = -3$$

$$\Rightarrow \cos(\angle ABC) = \frac{\vec{BA} \cdot \vec{BC}}{|\vec{BA}| |\vec{BC}|} = \frac{-3}{\sqrt{6} \sqrt{14}}$$

$$\angle ABC = 109.1^\circ$$

- 10** Points A and B are defined by the position vectors $a = i + 4j - 4k$ and $b = 2i + 5j - k$.
Let P be the point on OB such that AP is perpendicular to OB .
Then $\vec{OP} = q\vec{b}$, for a constant q .

- a** Express $\vec{AP}$ in terms of q , a and b .
- b** Use the fact that $\vec{AP} \cdot \vec{OB} = 0$ to find the value of q .
- c** Find the coordinates of the point P .



$$a) \quad \vec{AP} = \vec{AO} + \vec{OP} = -\vec{OA} + \vec{OP} = -a + qb$$

$$b) \quad \vec{AP} = -\vec{i} - 4\vec{j} + 4\vec{k} + q(2\vec{i} + 5\vec{j} - \vec{k})$$

$$= (2q-1)\vec{i} + (5q-4)\vec{j} + (4-q)\vec{k}$$

$$\vec{AP} \cdot \vec{OB} = 0 \Rightarrow 2(2q-1) + 5(5q-4) + (-1)(4-q) = 0$$

$$4q - 2 + 25q - 20 - 4 + q = 0$$

$$q = \frac{13}{15}$$

$$c) \quad \vec{OP} = q \cdot b = \frac{13}{15} (2\vec{i} + 5\vec{j} - \vec{k}) = \frac{26}{15}\vec{i} + \frac{13}{3}\vec{j} - \frac{13}{15}\vec{k}$$

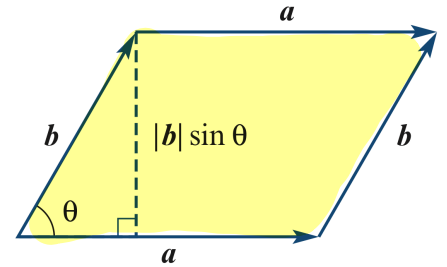
$$P\left(\frac{26}{15}, \frac{13}{3}, -\frac{13}{15}\right)$$

Vector Product (Cross Product)

Geometric definition

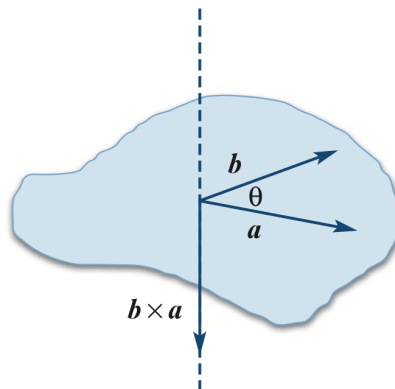
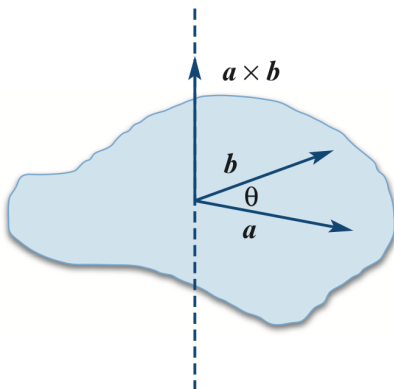
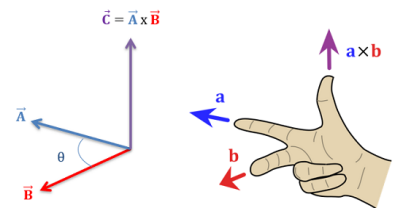
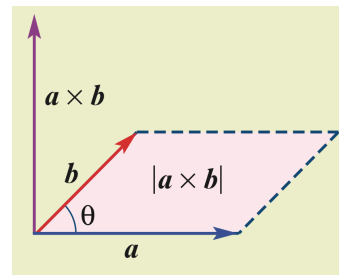
The vector product of $\vec{a}$ and $\vec{b}$, denoted as $\vec{a} \times \vec{b}$

- * The magnitude of $\vec{a} \times \vec{b}$ is equal to $|\vec{a}| |\vec{b}| \sin \theta$, θ is angle between $\vec{a}$ and $\vec{b}$. i.e. $|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$



$|\vec{a} \times \vec{b}|$ is the area of parallelogram 'spanned' by the vectors $\vec{a}$ and $\vec{b}$.

- * The direction of $\vec{a} \times \vec{b}$ is perpendicular to the plane containing $\vec{a}$ and $\vec{b}$, in the sense of right-hand screw turned from $\vec{a}$ to $\vec{b}$.



$$\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$$

$\vec{b} \times \vec{a}$ has the same magnitude as $\vec{a} \times \vec{b}$ but opposite direction.

Note : $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$$

Vector product of parallel vectors

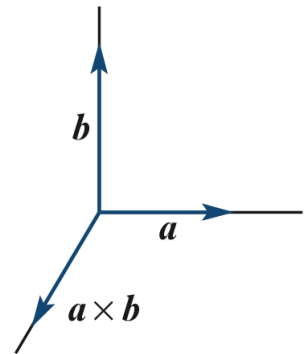
If $\vec{a}$ and $\vec{b}$ are parallel vectors, then $\vec{a} \times \vec{b} = 0$

$$\text{since } |\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin 0^\circ = 0$$

Vector product of perpendicular vectors

If $\vec{a}$ and $\vec{b}$ are perpendicular vectors, then

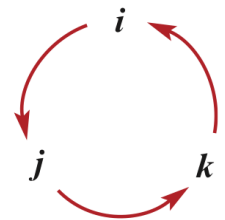
$$\begin{aligned} |\vec{a} \times \vec{b}| &= |\vec{a}| |\vec{b}| \sin 90^\circ \\ &= |\vec{a}| |\vec{b}| \end{aligned}$$



Vector product in component form

Using these observations about the vector product of parallel and perpendicular vectors:

- | | | |
|---------------------|---------------------|---------------------|
| ■ $i \times i = 0$ | ■ $j \times j = 0$ | ■ $k \times k = 0$ |
| ■ $i \times j = k$ | ■ $j \times k = i$ | ■ $k \times i = j$ |
| ■ $j \times i = -k$ | ■ $k \times j = -i$ | ■ $i \times k = -j$ |



$$\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$$

$$\text{If } \vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$$

$$\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$$

$$\text{then } \vec{a} \times \vec{b} = (a_2 b_3 - a_3 b_2) \hat{i} - (a_1 b_3 - a_3 b_1) \hat{j} + (a_1 b_2 - a_2 b_1) \hat{k}$$

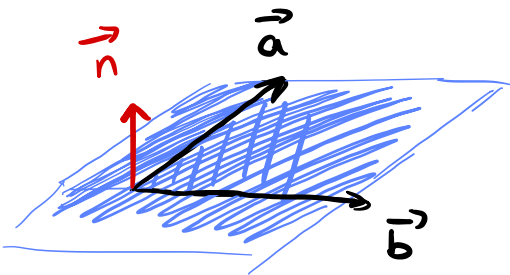
$$\vec{a} \times \vec{b} = \begin{vmatrix} \overset{+}{\hat{i}} & \overset{-}{\hat{j}} & \overset{+}{\hat{k}} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= + \hat{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \hat{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \hat{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

$$= (a_2 b_3 - a_3 b_2) \hat{i} - (a_1 b_3 - a_3 b_1) \hat{j} + (a_1 b_2 - a_2 b_1) \hat{k}$$

Example 18

Find the vector product of $\vec{a} = 3\hat{i} + 3\hat{j} + 8\hat{k}$ and $\vec{b} = \hat{i} - 3\hat{j} + 2\hat{k}$, and hence find a unit vector that is perpendicular to both $\vec{a}$ and $\vec{b}$.



$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 3 & 8 \\ 1 & -3 & 2 \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 8 \\ -3 & 2 \end{vmatrix} \hat{i} - \begin{vmatrix} 3 & 8 \\ 1 & 2 \end{vmatrix} \hat{j} + \begin{vmatrix} 3 & 3 \\ 1 & -3 \end{vmatrix} \hat{k}$$

$$= (3 \times 2 + 24) \hat{i} - (3 \times 2 - 8) \hat{j} + (-9 - 3) \hat{k}$$

$$= 30 \hat{i} + 2 \hat{j} - 12 \hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{30^2 + 2^2 + 12^2} = 2\sqrt{262}$$

$$\text{unit vector} : \frac{1}{2\sqrt{262}} (30\hat{i} + 2\hat{j} - 12\hat{k})$$

Example 19

a Simplify:

i $a \times (a - b)$ ii $(a \times b) \cdot a$

b Given that $a \times b = c \times a$, with $a \neq 0$, show that $b = -c$ or $a = k(b + c)$ for some $k \in \mathbb{R}$.

Solution.

a. i. $\vec{a} \times (\vec{a} - \vec{b}) = \vec{a} \times \vec{a} - \vec{a} \times \vec{b} = -\vec{a} \times \vec{b}$

ii. $(\vec{a} \times \vec{b}) \cdot \vec{a} = 0$

$\vec{a} \times \vec{b}$ is perpendicular to $\vec{a}$

b. $\vec{a} \times \vec{b} = \vec{c} \times \vec{a} \quad \vec{a} \neq 0$

$$\vec{a} \times \vec{b} - \vec{c} \times \vec{a} = 0$$

$$\Downarrow \vec{a} \times \vec{c} = -\vec{c} \times \vec{a}$$

$$\vec{a} \times \vec{b} + \vec{a} \times \vec{c} = 0$$

$$\vec{a} \times (\vec{b} + \vec{c}) = 0$$

$\therefore \vec{a} \neq 0$ Two cases: $\begin{cases} \vec{a} \text{ and } \vec{b} + \vec{c} \text{ are parallel} \\ \vec{b} + \vec{c} = 0 \Rightarrow \vec{b} = -\vec{c} \end{cases}$

$\Rightarrow \vec{a} = k(\vec{b} + \vec{c}), \quad k \in \mathbb{R}$

$\downarrow$
magnitude.